

2/3/24

Casting

Gate ME : 1 to 2 M.
 PI : 3 to 4 M
 ESE pre : 1-2 Questions
 Main : 1-2 Questions

Steps in casting process

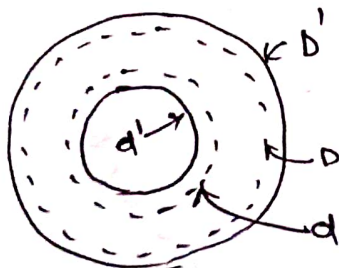
1. pattern Making
2. Mould and Core Making
3. pouring and Solidification
4. fettling
5. Inspection

Allowances provided to pattern

1. shrinkage
 2. Machining
- } Major allowances
3. draft
 4. shake
 5. distortion
- } minor allowances

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1. Shrinkage allowance



Hollow Casting

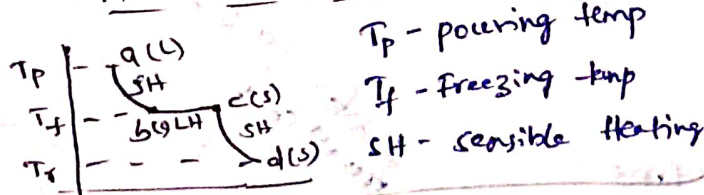
$$D' = D + \text{allowance}$$

$$d' = d - \text{allowance}$$

D = outer dia of Casting

d = Inner dia of Casting

Stages in Cooling in Casting



- Shrinkage allowance for compensate the shrinkage when the temp of solid phase drop from freezing to room temp
- room temp cast $\xrightarrow{\text{shrinkage allowance}}$ Indpt of lig metal

9359.455955

Zavi Sir
 glow ga chepa
 sir
 Maharashtra sir

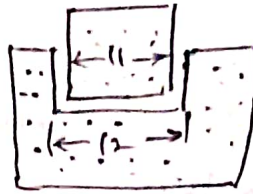
There diagram
 is v imp

Additional topics

1. Riser design
2. Special Casting process
3. defects

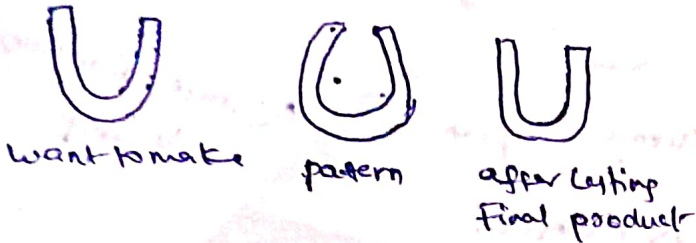
Draft allowance (or) taper allowance:

Shake allowance (or) Rapping allowance



Distortion allowance

properties of mould sand



1. porosity - ability to escape air

2

Net Buoyancy force = Buoyant force - Weight of core
(Weight of molten metal displaced) (W_c)

Chaplets are used to prevent the movement of core

$$\text{Net B.F} = V \cdot g \cdot (\rho_m - \rho_c)$$

Volume of core
gravity
molten metal density
Core density

(W_m)

Pouring time (or) Filling time = $\frac{\text{Volume}}{\text{Area} \times \text{velocity}}$

Ratio: $A_s : A_r : A_g$
 ↓
 cl area of sprue
 area of runner
 cl area of ingate

✓ $P.T = \frac{\text{Volume}}{A_{\min} \times V_{\max}}$

$\min(A_s, A_r, A_g)$

All units in mm

✓ $V_{\max} = \sqrt{2gh_{\text{sprue}}}$

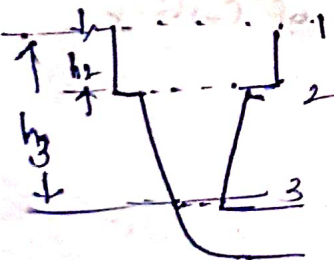
Pouring height also

$Re = \frac{\rho V D_s}{\mu} = \frac{V D_s}{\nu}$

Reed for laminar flow

Velocity
→ Sprue dia

→ Kinematic viscosity



Flow rate = $A_2 \times \sqrt{2gh_2} = A_3 \times \sqrt{2gh_3}$

$A_{\min} \times V_{\max}$

Solidification time

$\tau = K M^{\frac{n}{2}}$

time (sec) ← mould const

mould modulus

⇒ $\tau = K \left(\frac{V}{A_s} \right)^2$

Volume
Surf area

$\frac{\tau_1}{\tau_2} = \left(\frac{M_1}{M_2} \right)^2$

surf area
 $A_s = 2 \{ (400 \times 200) + (200 \times 140) + (400 \times 140) \}$

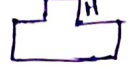
c/s area of cylinder = πab

surf .. = $\pi \sqrt{2(a^2 + b^2)}$

Heat removal rate $\overset{\text{FW}}{\underset{\text{mass}}{\overset{\text{latent heat}}{Q}}} = m \cdot L_H \Rightarrow L_H = \frac{Q \times \text{time}}{m}$ J/kg

Riser design

Side riser $H = D$  *note: if not mentioned in Q/b*

top riser $H = D/2$ 

Freezing ratio = $\frac{\text{Modulus of riser}}{\text{Modulus of mould}}$

$F.R = \frac{M_R}{M_C}$

$\frac{V}{A_s} = M_R = \text{Modulus of riser}$

Top cylinder riser

$\left[\frac{D}{6} = M_R \right] \Rightarrow \frac{V}{A_s} = \frac{\pi D^2 H}{\pi D H + \pi D^2}$

Side cylindrical riser

1. Caine's method

2. modulus method

3. Novel research method

$M_R = \frac{\frac{\pi D^2 H}{4}}{\pi D H + 2 \frac{\pi D^2}{4}}$

$M_R = \frac{D}{6}$

$M_R = 1.2 M_C$

$\frac{T_R}{T_C} = \left(\frac{M_R}{M_C} \right)^2$
 $\downarrow$
 Gelling solidification time $\checkmark$ Jungs

Blow Holes - poor permeability

misrun - Insufficient fluidity

Hot tearing - poor collapsibility

Wash - mold erosion

Welding (2-3 m)

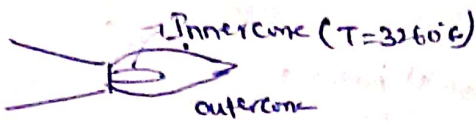
Fusion: parent metal melts.

Non-Fusion: without melting parent metal

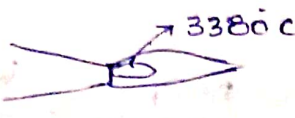
Gas welding

oxy acetylene gas welding

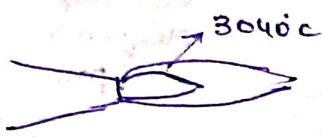
Neutral flame



oxidizing flame



carburizing flame



Ar₂ welding

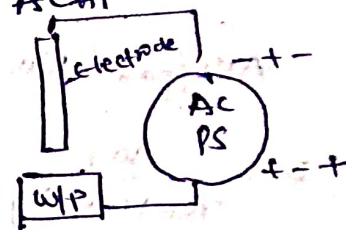
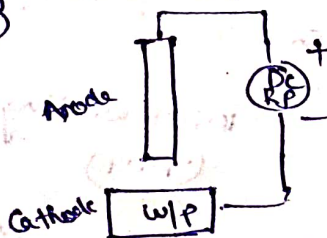
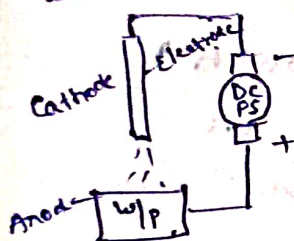
Direct current straight polarity

Alternating current High freq

DCSP (Direct current power supply)

DCRP

AC HF



Electrode touch w/p → short circuit

Electrode Not touch w/p → open circuit

Stable Equilibrium Cond

$$a + bL = V_0 - \frac{V_0}{I_s} \times I$$

→ open circuit voltage

→ short circuit voltage

$$V_a = a + bL$$

Optimum power
 $P = V_a \times I$

Stable Eg11

$$V_a = V_b$$

Rated and desired duty cycle:

$$I_s D_s = I_d D_d$$

I_s = rated current

D_s = rated D.C

I_d = desired current

D_d = desired D.C

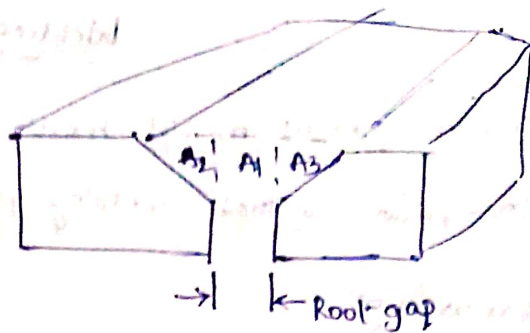
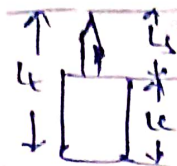
Duty cycle D.C = $\frac{\text{Arc on time (AOT)}}{\text{Total welding time}}$

$$D.C = \frac{AOT}{AOT + rest time}$$

Area of weld = $A_1 + 2A_2$
bead

Vol of weld (W.B) = $A \times \text{length of weld bead}$

Vol/electrode = $\frac{\pi}{4} d^2 \times L_e$



Total No. of electrodes required = $\frac{1.15 \text{ to } 1.25 \text{ Vol of weld bead}}{\text{Vol/electrode}}$

No. of passes = $\frac{\text{Total No. of electrodes required}}{\text{No. of electrodes/pass}}$

No. of electrodes required/pass = $\frac{\text{length of weld bead}}{\text{length of weld bead welded by one electrode}}$

Total welding time

Arc on time/pass (AOT) = $\frac{\text{length of weld bead}}{\text{Welding speed}}$

Total AOT = $\text{AOT/pass} \times \text{No. of passes}$

Duty cycle DC = $\frac{\text{AOT}}{\text{Total welding time}}$

Total welding time (TWT) = $\frac{\text{Total AOT}}{\text{D.C}}$

$\eta = \frac{\text{heat required (H.R)}}{\text{Heat generated (H.G)}}$

Heat req (H.R) = $m \times H.R/m$

$H.R = \rho \times \text{Vol} \times H.R/m$

Heat gen (H.G) = $\text{power} \times \text{time}$

Welding time = $\frac{\text{distance}}{\text{time}}$

Heat transfer efficiency $\eta = \frac{H/T}{VI}$

Sp melting energy $U = \rho (SH + CH)$
 $= \rho (C_p (T_m - T_0) + \Delta T_m)$
 Sensible heat, Latent heat, ambient temp

Melting efficiency $\eta_m = \frac{U A \eta}{\rho V I}$
 weld zone area

$$\frac{\text{Heat Input}}{\text{Length of weld}} = \frac{\eta \cdot V \cdot I}{l}$$

$$\text{Area of electrode} \times \text{feed rate} = \text{Area of weld} \times \text{Speed}$$

Resistance welding

Heat gen in

resistance welding

$$H \cdot E = I^2 R \cdot \tau$$

resistance
current flow
→ time

$$200 \text{ (microhm)}$$

$$\downarrow$$

$$200 \times 10^{-6} \text{ ohm}$$

$$\eta_{\text{melting}} = \frac{H \cdot R}{H \cdot E} \rightarrow \text{Heat required}$$

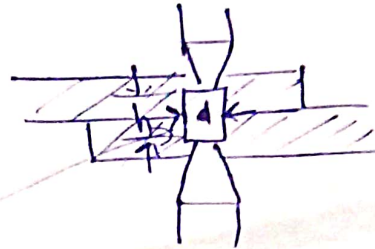
Heat gen →

$$H \cdot R = m (S_H + L_H)$$

$$= \rho \cdot V (S_H + L_H)$$

$$H \cdot R = \rho \times \left(\frac{\pi}{4} d^2 \times h \right) \left[\frac{S_H}{S} + \frac{L_H}{L} \right]$$

Sheet
ρ



d = nugget diameter
 h = nugget thickness

not given 100%
efficiency

Heat generated

$$(H \cdot E) = \text{Heat req} + \text{Heat dissipated}$$

Metal forming

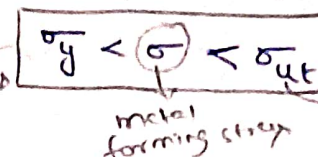
Metal forming: making desired shape of comp by generating plastic def in material with mechanical loads.

Cold Working

- Working below recrystallization temp
- Strength & Hardness $\uparrow\uparrow$
- b/c of strain hardening effect
- Grain structure is changed
- Scales and oxides not formed
- Surf finish is good
- coeff friction (μ) $\downarrow$ is 0.1 to 0.2
- Force req $\uparrow$ • dimension accuracy $\uparrow$

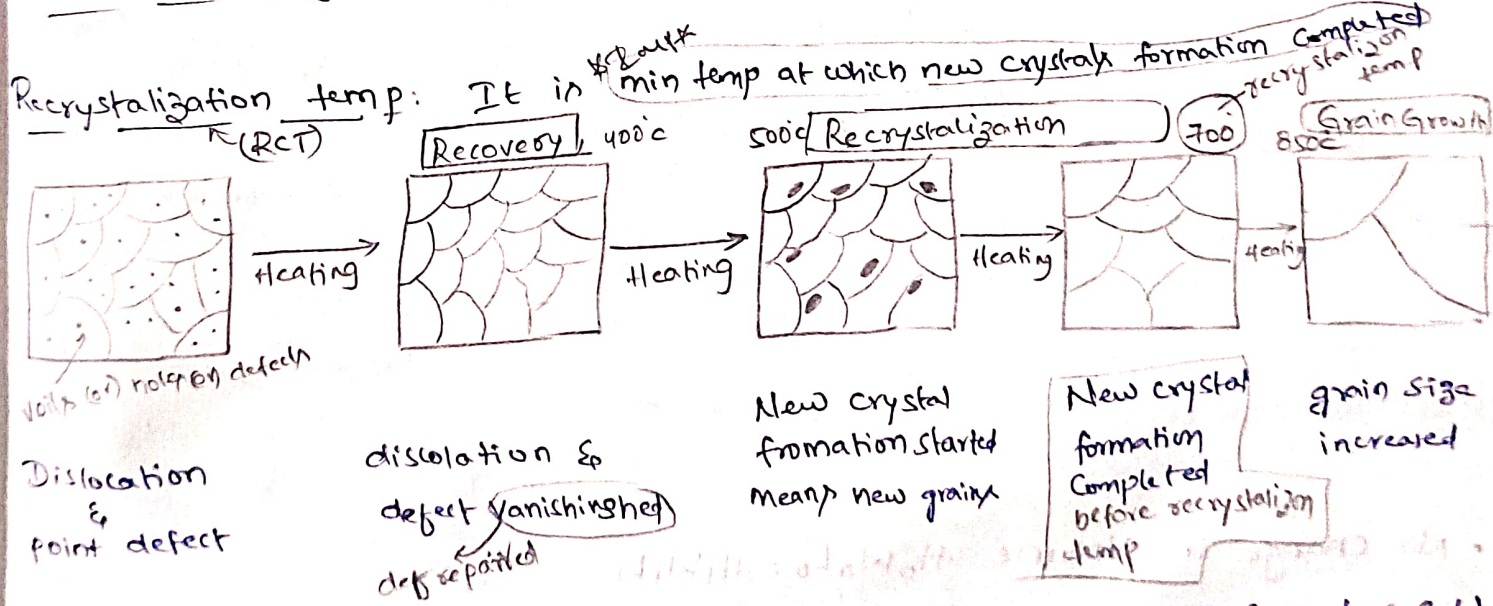
Hot Working

- Working above recrystallization temp
- ductility & toughness $\uparrow\uparrow$
- b/c grain flow without dislocation
- Scales and oxides formed (out side temp effect as billet temp is 500 so black colour on form is seen)
- Surf finish is rough
- $\mu \uparrow \rightarrow 0.4$ to 0.6
- Force req $\downarrow$ • dimension accuracy $\downarrow$



Warm working: heated just below recrystallization temp (R.C) to get adv of both Cold and hot working

Isothermal Working: Working at const temp by heating rollers also

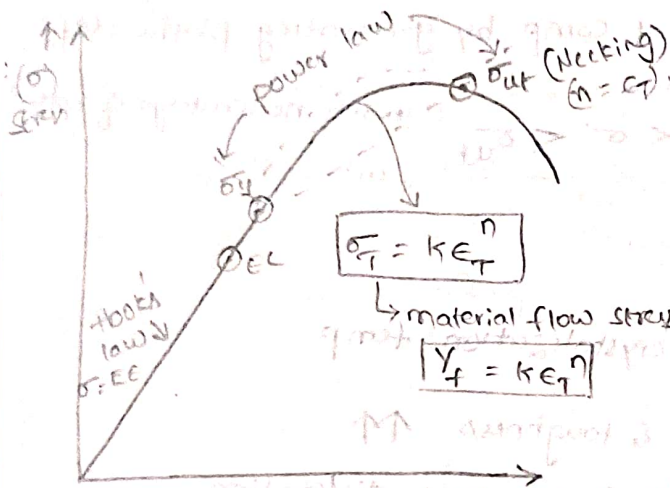


Recrystallization temp after Cold working: RCT is less than normal comp for Cold worked comp due to residual stresses and internal energy stored in mat left heat req - small

RCT after ^{hot} working: RCT is same before and after Hot working due to no internal energy and residual stress

grain size
 $\rho = \frac{1}{A} \ln \frac{V_0}{V}$
 $\rho = \frac{1}{A} \ln \frac{V_0}{V}$

power law of strain hardening: True stress = $\frac{\text{Load}}{\text{Inst area}}$



$$\sigma_T = \sigma_E (1 + \epsilon_E)$$

$$\sigma_T = \sigma_E \times \left(\frac{A_i}{A_f} \right)$$

$$= \sigma_E \times \left(\frac{d_i}{d_f} \right)^2$$

$$\sigma_T = \sigma_E \times \frac{r_f}{r_i}$$

$\gamma_f = \text{flow stress (or) True stress}$
 $k = \text{Strain hardening coeff}$
 $\epsilon_T = \text{True Strain}$
 $n = \text{Strain hardening exp}$
 $(0 < n < 1)$

$\epsilon_T = \text{True strain}$

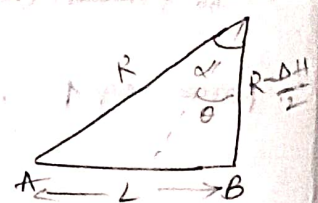
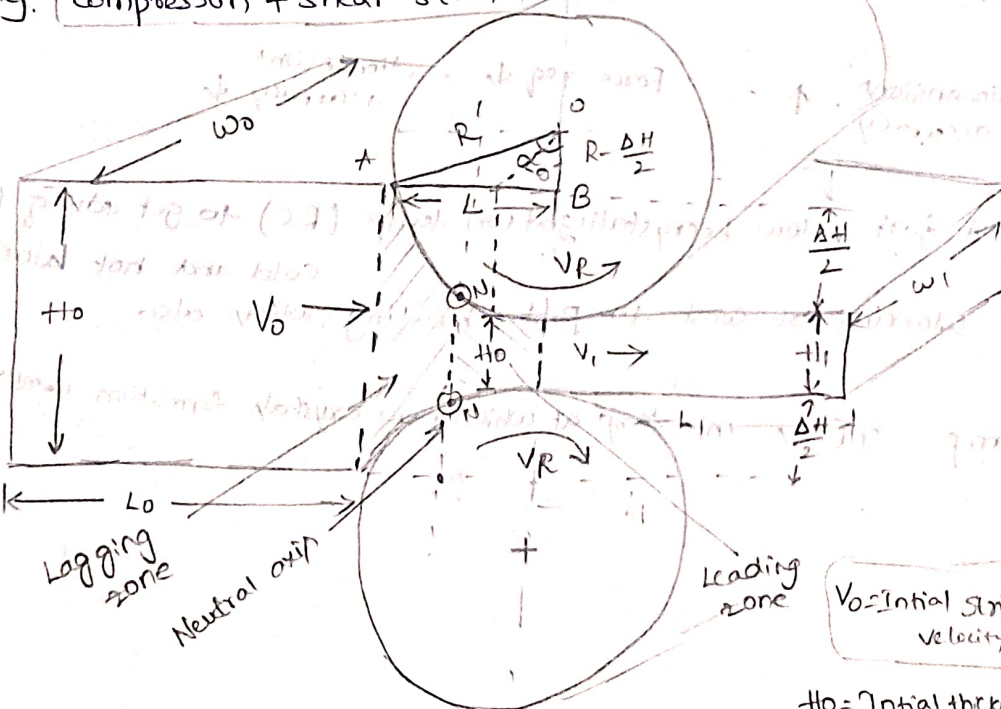
$$\epsilon_T = \ln \left(\frac{A_i}{A_f} \right) \checkmark$$

$$= 2 \ln \left(\frac{d_i}{d_f} \right)$$

$$\epsilon_T = \ln \left(\frac{r_f}{r_i} \right) \checkmark \quad \epsilon_T = \ln(1 + \epsilon_E)$$

Avg flow stress $(\gamma_f) = \frac{k \epsilon_T^n}{n+1}$ Flow stress $\sigma_T = k \epsilon_T^n \checkmark$

Rolling: Compression + shear stress



$L = \text{length of deformation}$

$R = \text{Radius of roller}$

$V_R = \text{Roller velocity}$

$V_s = \text{strip velocity}$

$N = \text{Roller speed (rpm)}$

$\alpha = \text{def angle (or) bite angle}$

$V_0 = \text{Initial strip velocity}$

$V_1 = \text{Final strip velocity}$

$H(\theta) = \text{Thickness of strip at an angle of } \theta \text{ from exit}$

$H_0 = \text{Initial thickness}$ $H_f = \text{Final thickness}$

$W_0 = \text{Initial width}$ $W_f = \text{Final width}$

$L_0 = \text{Initial length of strip}$ $L_f = \text{Final length of strip}$

No change in Volume $H_0 W_0 L_0 = H_f W_f L_f$

Elongation factor $= \frac{L_f}{L_0} = \frac{A_0}{A_f} = \frac{H_0 W_0}{H_f W_f}$

Volume flow rate discharge = Area x Velocity = C

$$A \times V = C$$

$V_0 = \text{Initial strip velocity}$

$$A_0 V_0 = A_f V_f \rightarrow V_f$$

$V_f = \text{final " "}$

$$H_0 W_0 V_0 = H_f W_f V_f$$

→ change in thickness

$\Delta H = H_0 - H_1$
 $\Delta H = D(1 - \cos \alpha)$

roller dia
 roller radius
 bite angle

$\tan \alpha = \frac{\Delta H}{R}$

→ Max reduction possible per pass in rolling

$\Delta H = H_0 - H_1 = \mu^2 R$

→ Length of deformation

$L = \sqrt{R(\Delta H)}$

→ change in length at angle θ from exit

$H_0 - H_1 = R\theta^2$

→ Backward (or) lagging zone

$V_s < V_R$

Slip = $\frac{V_s - V_R}{V_R} = -ve \Rightarrow \frac{V_0 - V_R}{V_R} \times 100$

pressure ↑ up to Neutral plane

→ Forward (or) leading zone

$V_s > V_R$

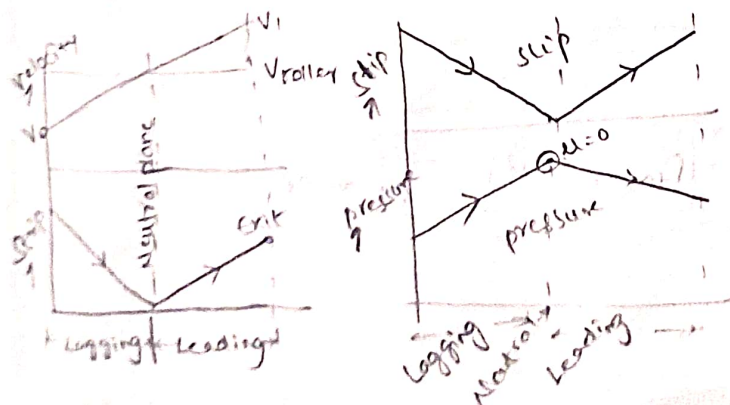
Slip = $\frac{V_s - V_R}{V_R} = +ve \Rightarrow \frac{V_1 - V_R}{V_R} \times 100$

pressure ↓ from Neutral plane

→ Neutral plane ($V_s = V_R$) = $\frac{\pi D N}{60}$

Slip = 0

Pressure is max at Neutral axis



Rolling Analysis friction coeff

$\mu \geq \tan \alpha$

Imp condition for rolling

$\mu = \tan \beta$

$\tan \beta \geq \tan \alpha$

friction angle $\beta \geq \alpha$ deformation angle

$\Delta H \propto \alpha \uparrow$

$\Delta H \propto \mu$ deformation

$\Delta H \propto R$

In this condition $\alpha = \beta$ - friction angle

$\mu = \tan \beta$

No. of passes req in rolling operation

Case-① Const draft (red in thickness)

draft $\Delta H = H_0 - H_1 = H_1 - H_2$

Min No. of passes

$n_{min} = \frac{H_0 - H_f}{\mu^2 R}$

max possible ΔH

Case ②

Elongation factor (k) const

$k = \frac{L_0}{L_1} = \frac{A_0}{A_1} \Rightarrow \frac{L_1}{L_0} = \frac{1}{k} \quad (\delta) \quad \frac{A_0}{A_1} = \frac{A_1}{A_2}$

$\frac{A_0}{A_n} = k^n = \frac{L_1}{L_0}$

Case ③

Const reduction in each pass

reduction (r) = $\frac{H_0 - H_1}{H_0} = \frac{H_1 - H_2}{H_1}$

$H_n = (1 - r)^n H_0$

reduction value (0.02, 0.3)

Plane strain condition $W_0 = W_1$

In problem not given two with thick b = W_1

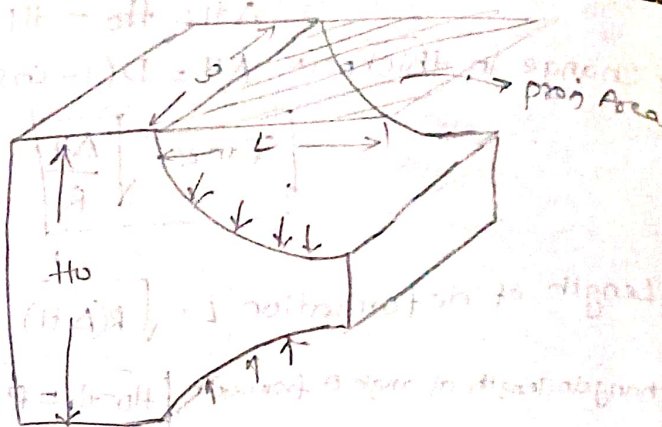
Rolling pressure (P_{avg})

Case-1: The avg roll pressure is given (P_{avg})

$$\text{Rolling force} = F_R = P_{avg} \times A_p$$

$$F_R = P_{avg} \times L \times W$$

$$F_R = P_{avg} \times \sqrt{R \Delta H} \times W$$



Case-2 Ideal rolling ($\mu=0$)

Condition A: The strip material is rigid and plastic

Condition B: Strip material is rigid and strain hardened

$$P_{avg} = \frac{\sigma_1 + \sigma_2}{2} = \frac{2}{\sqrt{3}} \sigma_y$$

$$F_R = \frac{2}{\sqrt{3}} \sigma_y \cdot L \cdot W$$

$$\text{Rolling force} = F_R = \frac{2}{\sqrt{3}} \bar{\sigma}_f \cdot L \cdot W$$

$$\text{Avg. flow stress } \bar{\sigma}_f = \frac{K \epsilon^n}{n+1}$$

Case-3: Considering friction b/w roll and work piece ($\mu \neq 0$) (This for

Condition A: The strip material is rigid and plastic (σ_y)

Condition B: The strip material is rigid and strain hardened ($\bar{\sigma}_f$)

$$F_R = \frac{2}{\sqrt{3}} \sigma_y \left(1 + \frac{\mu L}{2 H_{avg}} \right) \times L \times W$$

$$F_R = \frac{2}{\sqrt{3}} \bar{\sigma}_f \left(1 + \frac{\mu L}{2 H_{avg}} \right) \times L \times W$$

$$H_{avg} = \frac{H_0 + H_1}{2}$$

Power required in rolling Total power = $P_{total} = 2 \text{ power per roller}$

$$\text{Power per roller} = \text{Torque} \times \text{angular speed}$$

$$= T \times \omega$$

$$\omega = \frac{V_1 / 60}{R} \text{ rad/s}$$

$$\text{Power per roller} = F_R \times \frac{L}{2} \times \frac{2\pi N}{60}$$

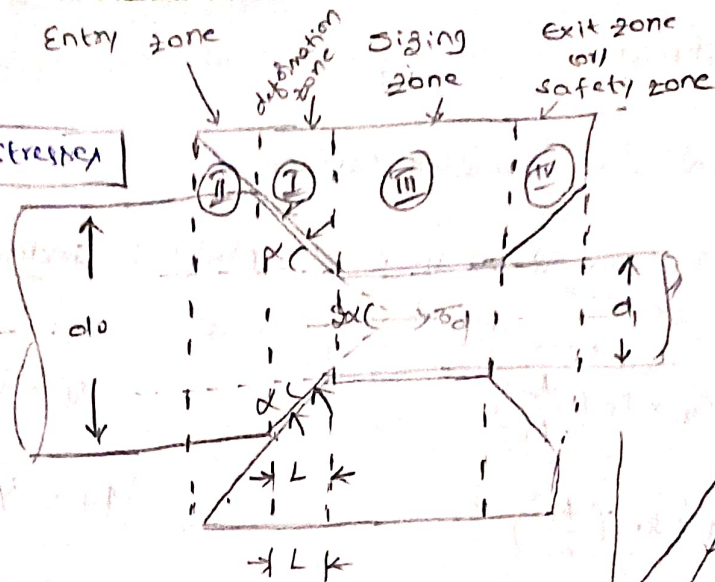
Wire drawing

Tensile + compression + shear stresses

2α = full die angle

α = semi/half die angle

L = length of deformation



II : Entry zone (or) lubricating zone:

I : Deformation zone

III : Sizing zone $\rightarrow$ converting elastic deformation into plastic deformation

IV : exit (or) safety zone $\rightarrow$ collect lubricants ~~and~~ to avoid formation of surface cracks.

Deformation zone

Case I : Const length deformation
reduction in cross section area increase

by increasing die angle but due to sudden change in c/s area more drawing stress

and σ_{eq} Some time wire will break in def zone

So large die angle are not preferable

Case - II $\alpha_3 < \alpha_1 < \alpha_2$ Const red in c/s area

The length of def zone Increase by reduction die

angle

Preparation of work

1. Annealing : heat and cool the wire in furnace

2. cleaning : i. Acid pickling : Removing chemicals, corrosion from surf of wire

ii. Alkaline cleaning : Removing oil, grease from surf of wire

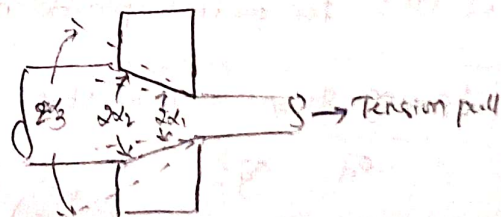
iii. phosphate cleaning : To avoid direct cont with die & wire

iv. Lubrication : To reduce friction & dissipate heat

3. Pointing : making taper at end

II Entry (or) lubricating zone

Lubricant	Colour	Effect
Gaseous lubricant	Black/brown	↑ Heat ↑ burning
liq. lubri	metallic/dull	↓ Heat ↓
Solid powder lubricant	Glittery or silver	microcracks on surfaces



Drawing stress

Case ①: Ideal wire drawing ($\mu=0$)

Condition (A): The wire metal is rigid & plastic

$$\frac{F_d}{A_1} = \sigma_d = \sigma_y \times \ln\left(\frac{A_0}{A_1}\right)$$

$$F_d = \sigma_y \times A_1 \times \ln\left(\frac{A_0}{A_1}\right)$$

Condition (B): Wire metal is rigid & strain hardened

$$\sigma_d = \bar{\sigma}_f \times \ln\left(\frac{A_0}{A_1}\right)$$

$$F_d = \bar{\sigma}_f \times A_1 \times \ln\left(\frac{A_0}{A_1}\right)$$

Case ②: Actual wire drawing ($\mu \neq 0$) $\rightarrow$ No need to remember formula given in gate

frictional factor $B = \frac{\mu}{\tan \alpha} = \mu \cot \alpha$

Cond (A): Wire is rigid & plastic

$$\sigma_d = \sigma_y \times \left(\frac{1+B}{B}\right) \left[1 - \left(\frac{A_1}{A_0}\right)^B\right]$$

$$F_d = \sigma_y \times A_1 \times \left(\frac{1+B}{B}\right) \left[1 - \left(\frac{A_1}{A_0}\right)^B\right]$$

Cond (B): Wire is rigid and strain hardened

$$\sigma_d = \bar{\sigma}_f \times \left(\frac{1+B}{B}\right) \left[1 - \left(\frac{A_1}{A_0}\right)^B\right]$$

$$F_d = \bar{\sigma}_f \times A_1 \times \left(\frac{1+B}{B}\right) \left[1 - \left(\frac{A_1}{A_0}\right)^B\right]$$

Total drawing stress (Incl die cond)

$$\sigma_{\text{Total}} = \sigma_y + (\sigma_d - \sigma_y) e^{-\frac{2\mu L}{r_1}}$$

not imp for gate

Max percentage reduction

$$\% \text{ Reduction} = \frac{A_0 - A_1}{A_0} \times 100$$

Case ①: max percentage reduction ideal wire drawing ($\mu=0$)

Cond (A): For perfect plastic material

Cond (B): For strain hardened material

Drawing stress = Yield stress

$$\sigma_d = \sigma_y$$

$$\sigma_y \ln\left(\frac{A_0}{A_1}\right) = \sigma_y$$

$$\ln\left(\frac{A_0}{A_1}\right) = 1$$

$$\frac{A_0}{A_1} = e^1$$

$$\text{Max. \% reduction} = \frac{A_0 - A_1}{A_0} \times 100 = \left(1 - \frac{A_1}{A_0}\right) \times 100$$

$$= (1 - e^{-1}) \times 100 \approx 63.2\%$$

$$\sigma_d = \bar{\sigma}_f$$

$$\bar{\sigma}_f \ln\left(\frac{A_0}{A_1}\right) = \bar{\sigma}_f$$

$$\frac{K \epsilon_f^n}{n+1} \ln\left(\frac{A_0}{A_1}\right) = \frac{K \epsilon_f^n}{n+1}$$

$$\ln\left(\frac{A_0}{A_1}\right) = n+1 \rightarrow \frac{A_0}{A_1} = e^{n+1}$$

$$\text{Max \% red} = \frac{A_0 - A_1}{A_0} \times 100$$

case ②: max percentage red per pass in actual wire drawing ($\mu \neq 0$)

cond (A) : rigid and plastic

$$\sigma_d = \sigma_y$$

$$\sigma_y \left(\frac{1+B}{B} \right) \left[1 - \left(\frac{A_1}{A_0} \right)^B \right] = \sigma_y$$

$$1 - \left(\frac{A_1}{A_0} \right)^B = \frac{B}{1+B}$$

$$\left(\frac{A_1}{A_0} \right) = \left(\frac{1}{1+B} \right)^{1/B}$$

$$\text{Max \% red} = \left(1 - \frac{A_1}{A_0} \right) \times 100 = \left[1 - \left(\frac{1}{1+B} \right)^{1/B} \right] \times 100$$

cond (B) rigid and strain hardened

$$\sigma_d = \sigma_y$$

$$Y_f \left(\frac{1+B}{B} \right) \left[1 - \left(\frac{A_1}{A_0} \right)^B \right] = Y_f$$

$$\frac{k \epsilon_1^n}{n+1} \left(\frac{1+B}{B} \right) \left[1 - \left(\frac{A_1}{A_0} \right)^B \right] = \frac{k \epsilon_1^n}{n+1}$$

$$\frac{A_1}{A_0} = \left[1 - (n+1) \left(\frac{B}{1+B} \right) \right]^{1/B}$$

$$\text{Max \% red} = \left(1 - \frac{A_1}{A_0} \right) \times 100$$

$$\Rightarrow \left[1 - \left[1 - (n+1) \left(\frac{B}{1+B} \right) \right]^{1/B} \right]$$

Max possible % reduction in area per pass during an ideal plastic material with out friction is of order of 63.2%.

Tube drawing (In gate not asking ex)

Stress in tube drawing from this tube drawing

$$\text{Power req in tube drawing } P = F_{\text{tube}} \times V_{\text{exit}}$$

→ tube exit velocity

Case - ① Ideal tube drawing ($\mu = 0$)

cond (A) : The tube is rigid and plastic

$$\sigma_d = \sigma_y \ln \left(\frac{h_0}{h_1} \right)$$

$$F_d = \sigma_y \times A_1 \times \ln \left(\frac{h_0}{h_1} \right)$$

cond (B) material is rigid and strain hardened

$$\sigma_d = Y_f \ln \left(\frac{h_0}{h_1} \right)$$

$$F_d = Y_f \times A_1 \times \ln \left(\frac{h_0}{h_1} \right)$$

Case - ②: Actual tube drawing ($\mu_1 \neq \mu_2 \neq 0$)

$$\text{friction factor } B = \frac{\mu_1 + \mu_2}{\tan \alpha - \tan \beta}$$

cond (A) material rigid and plastic

$$\sigma_d = \sigma_y \left(\frac{1+B}{B} \right) \left[1 - \left(\frac{h_1}{h_0} \right)^B \right]$$

$$F_d = \sigma_y \times A_1 \times \left(\frac{1+B}{B} \right) \left[1 - \left(\frac{h_1}{h_0} \right)^B \right]$$

cond (B) : rigid and strain hardened

$$\sigma_d = Y_f \left(\frac{1+B}{B} \right) \left[1 - \left(\frac{h_1}{h_0} \right)^B \right]$$

$$F_d = Y_f \times A_1 \times \left(\frac{1+B}{B} \right) \left[1 - \left(\frac{h_1}{h_0} \right)^B \right]$$

Case ③ movable mandrel $\mu_1 = \mu_2 = \mu$

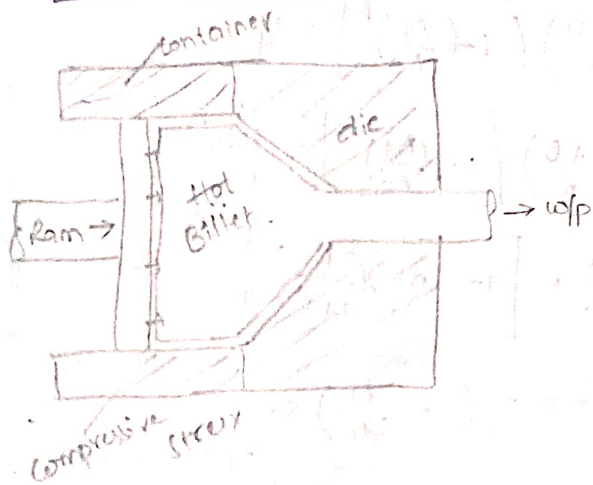
$$B = \frac{\mu_1 + \mu_2}{\tan \alpha} = 0$$

$$\sigma_d = \sigma_y \left(\frac{1+B}{B} \right) \left[1 - \left(\frac{h_1}{h_0} \right)^B \right] \quad \left(\frac{1+B}{B} \right) \left(\frac{1}{0} \right) \left(\frac{1}{0} \right) \Rightarrow \sigma_d = \sigma_y \ln \left(\frac{h_0}{h_1} \right) \quad \text{or } Y_f \ln \left(\frac{h_0}{h_1} \right)$$

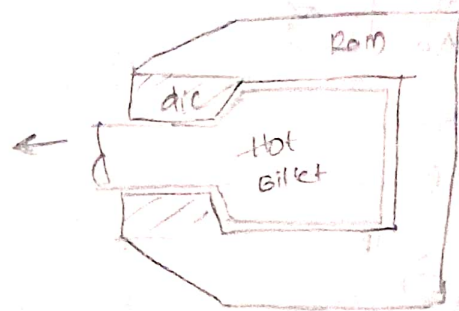
4. Extrusion :

Extrusion is more efficient than wire drawing for reduction in c/s area

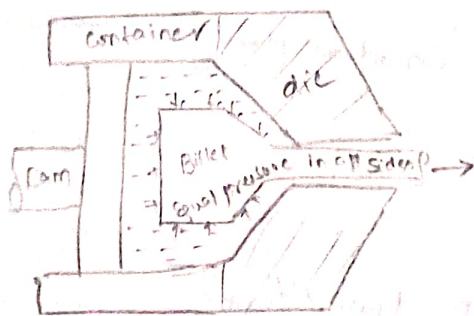
Forward or direct extrusion



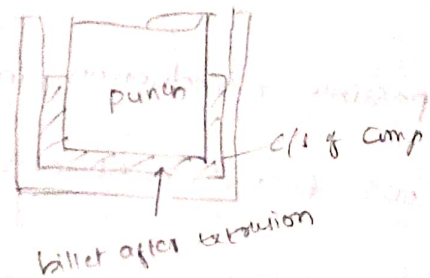
Backward or indirect extrusion



Hydrostatic extrusion



Impact extrusion



Ram pressure and extrusion load

$$\text{Power req in extrusion} = F_E \times V_{\text{ram}}$$

Case-1 Ideal extrusion ($\mu=0$)

Cond A: Billet is rigid and plastic

Extrusion stress $\sigma_E = \frac{F_d}{A_0} = \sigma_y \times \ln\left(\frac{A_0}{A_1}\right)$

Initial Area *Extrusion ratio*

$F_d = \sigma_y \times A_0 \times \ln\left(\frac{A_0}{A_1}\right)$

Initial area

Cond B Billet is rigid and strainhardened

$\bar{Y}_f = \frac{k \epsilon^n}{n+1}$ avg flow stress

Ram pressure $p = \bar{Y}_f \times \ln\left(\frac{A_0}{A_1}\right)$

$F_E = \bar{Y}_f \times A_0 \times \ln\left(\frac{A_0}{A_1}\right)$

Case-2 Indirect extrusion ($\mu_{\text{container}}=0, \mu_{\text{die}} \neq 0$)

$B = \mu \cot \alpha$

Ram pressure $P_{\text{ram}} = \sigma_y \left(\frac{1+B}{B} \right) \left[1 - \left(\frac{A_1}{A_0} \right)^B \right]$

$F_E = \sigma_y \times A_0 \times \left(\frac{1+B}{B} \right) \left(1 - \left(\frac{A_1}{A_0} \right)^B \right)$

Case 2, not at all Imp

Case ③ Direct extension ($L_{canl} \neq 0$, $L_{die} \neq 0$)

Ram pressure $P_r = \sigma_y \left(a + b \ln \left(\frac{A_0}{A_1} \right) + \frac{2L_0}{d_0} \right)$

extension force $F_E = \sigma_y \times A_0 \times \left(a + b \ln \left(\frac{A_0}{A_1} \right) \right)$

$F_E = \bar{\sigma}_f \times A_0 \times \left[a + b \ln \left(\frac{A_0}{A_1} \right) \right]$

Case ③ formula given in gate exam

Ideal case - Not given formula
Actual case : formula given

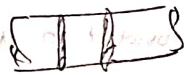
Case ④ Hot extension

Ram pressure $P_{ram} = K \ln \left(\frac{A_0}{A_1} \right)$

extension force $F_E = (K) \times A_0 \times \ln \left(\frac{A_0}{A_1} \right)$
 Extrusion Const (K) at temp t

Defects in extrusion

1. Bamboo defect



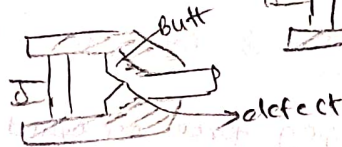
- formed due to non-uniform velocity

2. Central burst (or) pin hole defect



due to blow die and billet
Central cracks are generated

3. Fish tail defect



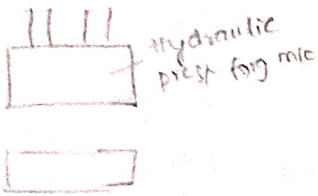
$\left(\frac{A_0}{A_1} \right) = \frac{P_r}{K}$

$\left(\frac{A_0}{A_1} \right) = \frac{P_r}{K}$

$\left(\frac{A_0}{A_1} \right) = \frac{P_r}{K}$

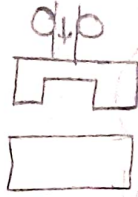
5. Forging

press forging



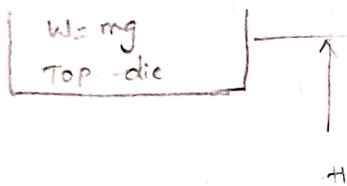
forging force applied gradually

drop hammer forging



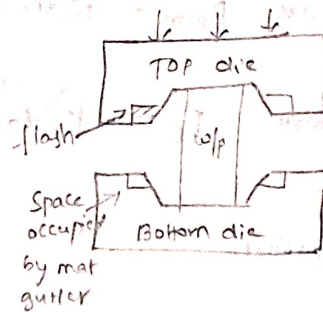
sudden forging force

Drop hammer forging

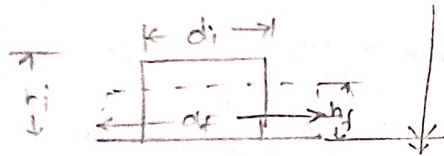


$$F_{avg} = \frac{W H}{h_i - h_f}$$

$$F_{avg} = \frac{(mg + PA) H}{h_i - h_f}$$



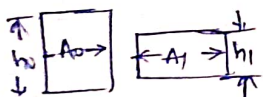
closed die forging



Gutter: free space provided in die to occupy excess volume of raw material
Flash: excess volume of material occupied in gutter

Analysis of forging

(Open die forging only & closed die not in syllabus)



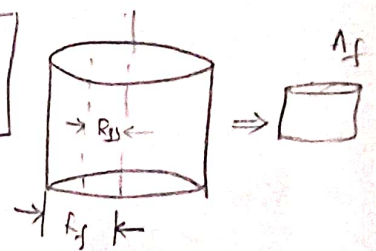
Volumetric change = 0
 $A_0 h_0 = A_f h_f$

$$\frac{\pi}{4} d_0^2 h_0 = \frac{\pi}{4} d_f^2 h_f$$

$$d_0^2 h_0 = d_f^2 h_f$$

Forging force on cylindrical body

$$R_{ss} = R_f - \frac{h_f}{2\mu} \ln\left(\frac{1}{\sqrt{3} \times \mu}\right)$$



Case ① Ideal Condition ($\mu = 0$)

Cond (A): w/p rigid and plastic

$$F_{max} = \sigma_y \times A_f$$

Case ② Actual forging ($\mu \neq 0$)

Cond A: w/p is rigid and plastic

$$F = \sigma_y \times A_f$$

Cond (B): w/p rigid and strain hardened

$$F = \gamma_f \cdot A_f = (K \epsilon_f^n) A_f$$

Cond (B) w/p rigid and strain hardened

$$F = \gamma_f \cdot A_f \left(1 + \frac{2\mu R_f}{3h_f}\right)$$

remember this two formulae in forging it is enough

forging operations

Fullering

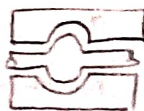


Drawing

upsetting

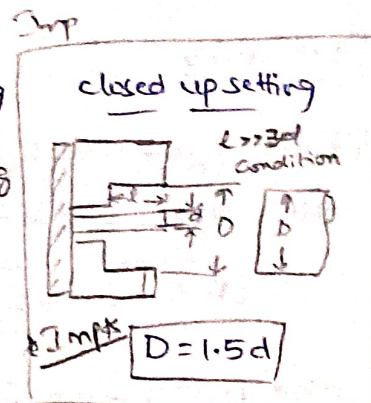


Edging



Blocking

Impression die forging



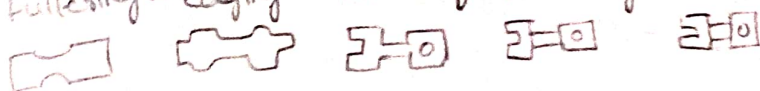
Problems Forging

• Energy = force $\times$ distance of wlp $\rightarrow$ change in height

• open die forging $\rightarrow$ Semi closed die $\rightarrow$ closed die

• connecting rod $\rightarrow$ Trimming or cutting

Fullering - edging - blocking - finishing - trimming



Extrusion

Tooth post tubes are produced by Impact extrusion

• Extrusion ratio = $\frac{A_0}{A_1}$

Rolling problems

1. $\tan \alpha = \sqrt{\frac{\Delta h}{R}}$ when we substitute and get value of α in degree only

$$\text{ex } \alpha = \tan^{-1} \sqrt{\frac{\Delta h}{R}}$$

degree only

But when we substitute we should substitute in rad only.

2. $T_u = 0.6 \sigma_u$

$$\sigma_y = 0.5 \sigma_u$$

3. $\sigma = 1400 \text{ MPa} \rightarrow \text{max load } \epsilon = 0.32 \frac{1}{3}$

Extrusion : long continuous metal

Forging : difficult parts (connecting rod, crank shaft)

metal spinning : spinning w/ metal turning axisymmetry
(Have hold large wheel)

Explosive welding : Boundary shearer corrosive resist to base metal

cladding reaction

Sheet Metal

Punching

Punch size = Hole size (H.S)

die size = H.S + clearance

$$D.S = P.S + 2C_r$$

Max punch force $F_{max} = \text{Ultimate Shear Strength} \times \text{Shear Area}$

$$F_{max} = \tau_u \times \pi d t \quad (d)$$

$$= \tau_u \times 40t \quad \square$$

$$= \tau_u \times 2(a+b)t \quad \square b$$

Energy required = $E = F_{max} \times \text{Penetration}$

Blanking

Die size (D.S) = Blank size

punch size (P.S) = D.S - clearance

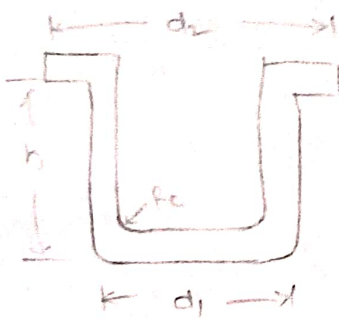
$$P.S = D.S - 2C_r$$

plate thickness $\geq 5\text{mm}$
Sheet = thickness $< 5\text{mm}$

$$\text{min punch dia} = d_{min} = \frac{F_{max}}{\frac{\pi}{4} d^2 \sigma_{Ac}}$$

$$d_{min} = \frac{4t\tau_u}{\sigma_{Ac}}$$

Flange Comp



$$1^{st} \text{ draw} = DRR = \frac{D_b - d_1}{D_b}$$

$$2^{nd} \text{ draw} = DRR = \frac{d_1 - d_2}{d_1}$$

$$3^{rd} \text{ draw} = DRR = \frac{d_2 - d_3}{d_2}$$

$$\frac{\pi}{4} D_b^2 = \frac{\pi}{4} d_1^2 + \pi d_1 h + \frac{\pi}{4} (d_2^2 - d_1^2) \quad \text{if } d_3 = 10.13 < 12$$

$$D_b = \sqrt{d_2^2 + 4d_1 h}$$

$$\text{drawing load force} = F_d = \pi D t \sigma_{ut}$$

Ex:

Bend radius stretch factor $1/2$
thickness

$$\text{Bend allowance} = (R + kt) \alpha_b \text{ radians}$$

$$= \left[100 + \left(\frac{1}{2} \times 2 \right) \right] \times 1$$

$$= 101 \text{ mm}$$

